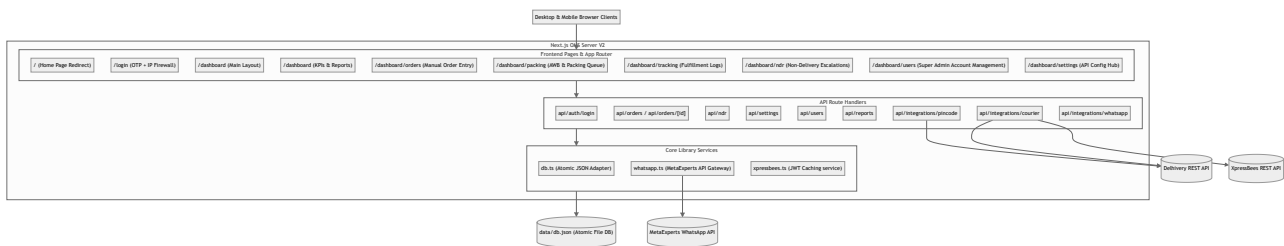


Technical Documentation: 99Store Order Management System (OMS)

Welcome to the technical and architectural documentation of the **99Store Order Management System (OMS)**. This web-based software is a premium, minimal dashboard application designed to streamline the complete lifecycle of order processing, packing, shipping agency routing, live dispatch tracking, WhatsApp customer automation, and Non-Delivery Report (NDR) case escalations.

1. System Architecture

The application is built using a modern full-stack TypeScript framework with a decoupled architecture. Below is a high-level architectural block diagram showing client, route handler, and data persistence connections:



2. Technology Stack & Directory Structure

The project utilizes the following core technology layers: 1. **Core Runtime:** [Next.js v16.2.6](#) and [React v19.2.4](#) with TypeScript. 2. **Styling:** Vanilla CSS structure via CSS variables and utility classes defined inside [globals.css](#), following a premium dark-themed minimalist aesthetics palette. 3. **Database:** Lightweight, thread-safe JSON-based atomic flat-file database located at `data/db.json`. 4. **Icons:** [lucide-react](#) vector graphics library.

Key File Mapping

- **System Layouts & Routing Rules:**
 - [globals.css](#): CSS variables for colors (Paid green, VIP amber, status tags), global reset, cards, premium inputs, buttons, tables, badges, modals, and responsive grids.
 - [layout.tsx](#): Root layout adding metadata headers and importing global styles.
 - [page.tsx](#): Entry checkpoint redirecting users based on active session storage.

- **Frontend Modules (Dashboard Area):**

- [login/page.tsx](#): OTP login entry containing quick-switch demo profiles, IP whitelist triggers, and local testing overrides.
- [dashboard/layout.tsx](#): App shell sidebar. Contains a **Reviewer Bar** at the top allowing developer/evaluator role switching on-the-fly to test view configurations.
- [dashboard/page.tsx](#): Status metrics, charts, courier performance graphs, and log export utilities.
- [dashboard/orders/page.tsx](#): Order Entry terminal with automatic pincode lookup and VIP tags.
- [dashboard/packing/page.tsx](#): Workspace to package orders and auto-assign courier AWBs.
- [dashboard/tracking/page.tsx](#): Delivery monitoring log showing raw courier API payloads.
- [dashboard/ndr/page.tsx](#): Non-Delivery Report dashboard containing tools to schedule delivery re-attempts.
- [dashboard/users/page.tsx](#): User details and role toggles.
- [dashboard/settings/page.tsx](#): Live API credential configurations and firewall switches.

- **Backend & Library Adapters:**

- [types.ts](#): Data contract model definitions.
- [db.ts](#): Atomically synchronized read/write functions for flat-file JSON management.
- [mockData.ts](#): Default seed datasets.
- [whatsapp.ts](#): MetaExperts WhatsApp message dispatcher.
- [xpressbees.ts](#): Auth token retriever with a 12-hour memory cache.

3. Database Schema

All system tables map to collections inside the flat file DB `data/db.json`. The records are managed through the database adapter methods in [db.ts](#).

3.1 Order Schema

Maps to the typescript interface [Order](#).

Field	Type	Description
<code>id</code>	<code>string</code>	Internal UUID

Field	Type	Description
orderId	string	Human-visible customer order code (e.g., 995-1001)
customerName	string	Full name of client recipient
phonePrimary	string	Main contact number (prefixed with 91 on dispatch API calls)
phoneSecondary	string (optional)	Alternate phone number
phoneTertiary	string (optional)	Alternate fallback phone number
address	string	Delivery physical address
pincode	string	6-digit destination postal code
state	string	Region state (mapped during pincode API fetch)
area	string	Local city / hub region
productDetails	string	Summary description of purchased products
paymentType	'COD' \ 'Paid'	Payment method. Generates green highlighters if Paid .
orderValue	number	Total invoice sum
weight	number	Order weight in kilograms
isVip	boolean	Flags a gold star next to the customer's name
status	OrderStatus	Current routing stage (see order flow list below)
awb	string (optional)	Courier Air Waybill tracker number
courier	'DTDC' \ 'XpressBees' \ 'Delhivery' \ 'Aggregator'	Chosen shipping carrier
eta	string (optional)	Estimated arrival date
createdBy	string	Creator account username

Field	Type	Description
history	OrderHistory[]	Transaction logs tracking state changes, editors, and remarks

3.2 User Schema

Maps to the typescript interface [User](#).

Field	Type	Description
id	string	Unique User UUID
username	string	Login username
name	string	Display name
role	UserRole	Operational clearance rank
isActive	boolean	Connection block toggle
lastLoginIp	string (optional)	Last registered login network address

3.3 NDR Record Schema

Maps to the typescript interface [NdrRecord](#).

Field	Type	Description
id	string	NDR case UUID
orderId	string	Connected order number
customerName	string	Customer name
phonePrimary	string	Contact phone
courier	string	Carrier name
awb	string	Waybill tracking code
reason	string	Delivery exception explanation from the carrier
status	'Pending' \ 'Re-attempt Scheduled' \ 'Returned to Origin'	Escalation status

Field	Type	Description
<code>retryDate</code>	<code>string</code> (optional)	Target date for courier retry
<code>internalNotes</code>	<code>string</code>	Escalation notes added by the tracking agent

4. User Access Controls (RBAC) & Security

The system employs **Role-Based Access Control (RBAC)** across frontend panels and endpoint systems. There are five clear [UserRole](#) types defined:

1. **Super Admin:** Holds full permissions. Accesses API integrations, user accounts, and database configurations.
2. **Order Team:** Permissions for manual order creation, editing customer details, and monitoring status reports.
3. **Packing Team:** Manages the packing queue, prints labels, selects couriers, and books shipments.
4. **Tracking Team:** Monitors transit statuses, tracks courier API payloads, and resolves NDR exceptions.
5. **Accounts Team:** Reviews reports, inspects COD collections vs. prepaid statuses, and verifies invoices.

4.1 UI Access Matrix

Restricted links are blocked at layout levels inside [dashboard/layout.tsx](#):

Route / Panel	Path Link	Super Admin	Order Team	Packing Team	Tracking Team	Accounts Team
Dashboard & KPI Reports	<code>/dashboard</code>	✓	✓	✓	✓	✓
Order Entry & Edit	<code>/dashboard/orders</code>	✓	✓	✗	✗	✓
Packing & Courier Select	<code>/dashboard/packing</code>	✓	✗	✓	✗	✗
Fulfillment & Dispatch	<code>/dashboard/tracking</code>	✓	✗	✗	✓	✗

Route / Panel	Path Link	Super Admin	Order Team	Packing Team	Tracking Team	Accounts Team
NDR & OFD Escalations	/dashboard/ndr	✓	✗	✗	✓	✗
User Management	/dashboard/users	✓	✗	✗	✗	✗
API Settings Hub	/dashboard/settings	✓	✗	✗	✗	✗

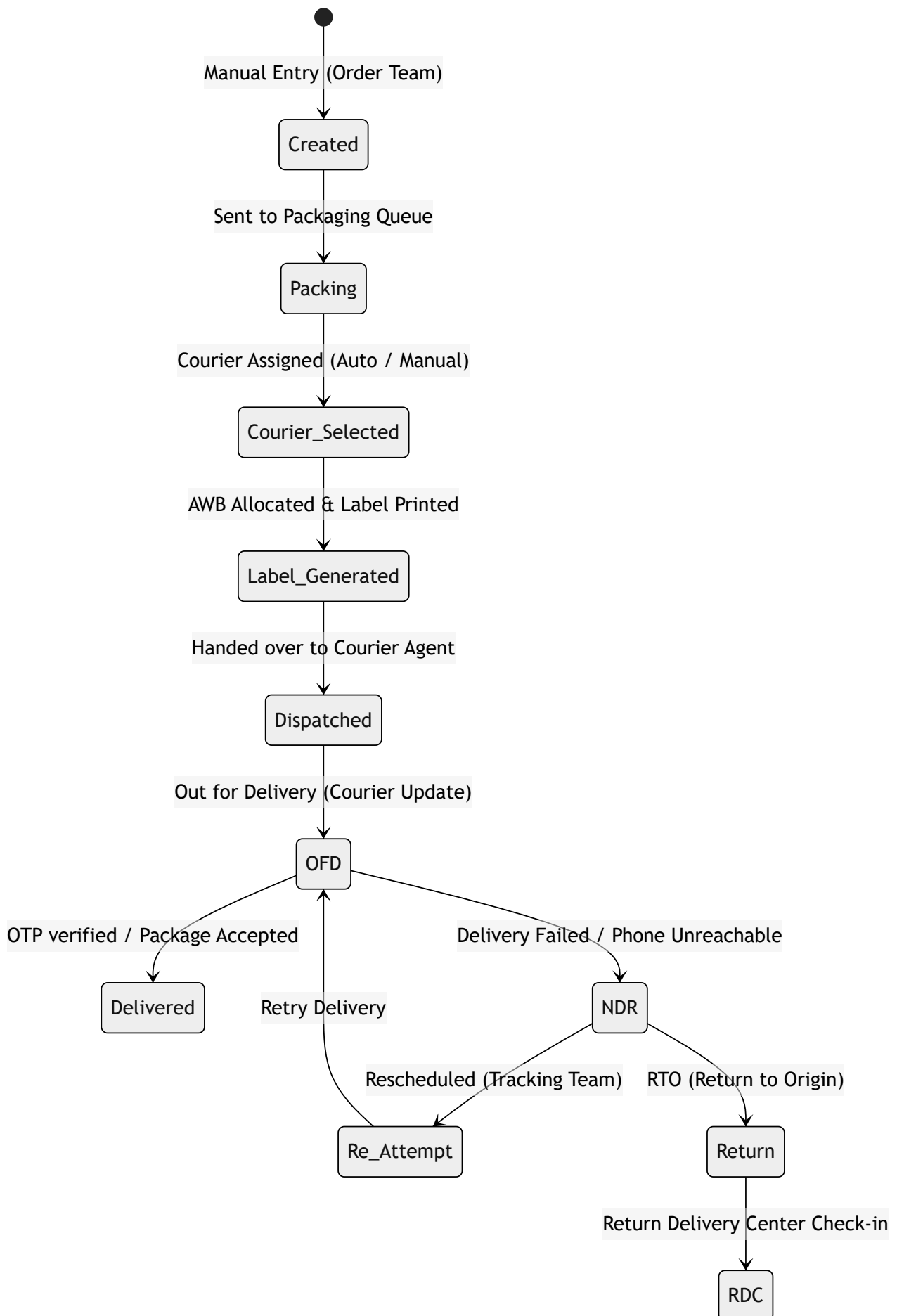
4.2 Security Firewall

- **IP Whitelist:** Enabled via [SystemSettings](#). When turned on, the system verifies `x-forwarded-for` or `x-real-ip` headers during login requests inside [route.ts](#). Non-whitelisted IPs receive an HTTP `403 Forbidden` response.
- **Demo Whitelist Bypass:** A client-side option is provided on the blocked modal to allow local reviewers and developers to easily bypass this check when testing locally.
- **OTP Verification:** The login flow requires a 6-digit OTP code. In the current development environment, the gateway accepts `999999` (or any valid 6-digit numeric pattern) for ease of demo navigation.

5. Main Functional Modules

5.1 Order Workflow

Orders pass through distinct operational milestones inside the OMS system:



5.2 Auto-Courier Selection logic

During the packing phase, if `autoCourierEnabled` is active in settings, the system executes an automated routing logic inside [packing/page.tsx](#) to identify the best shipment carrier: 1. **Weight-based Routing:** * Orders under **1.0 kg** default to **DTDC** (configured as highest priority for lightweight surface courier parcels). * Orders between **1.0 kg and 2.0 kg** default to **XpressBees**. * Orders exceeding **2.0 kg** default to **Delhivery** (or the Aggregator API). 2. **Config Priority Override:** If specific carriers are disabled in settings, the auto-router shifts down the active list using the configured priorities.

5.3 Pincode Area Auto-Fetcher

When an Order Team agent inputs a pincode in the Order Entry form, an API call queries [api/integrations/pincode](#): 1. **Live Check:** If Delhivery is active and has a configured API Key, the endpoint calls Delhivery's live serviceability database (`/c/api/pin-codes/json/?filter_codes=`). 2. **Internal Database:** If live serviceability is offline, it checks the local mapping index (`pincodeMap`). 3. **Zone Fallback:** If the pincode is completely new, it automatically parses the first digit (postal zone) and resolves a fallback region: * `1xxxx` $\rightarrow$ Delhi/NCR * `2xxxx` $\rightarrow$ Uttar Pradesh * `3xxxx` $\rightarrow$ Gujarat * `4xxxx` $\rightarrow$ Maharashtra * `5xxxx` $\rightarrow$ Karnataka/Andhra Pradesh * `6xxxx` $\rightarrow$ Tamil Nadu/Kerala * `7xxxx` $\rightarrow$ West Bengal/NorthEast * `8xxxx` $\rightarrow$ Bihar/Jharkhand

5.4 WhatsApp Notification Gateway

Integrated via [whatsapp.ts](#), status transitions trigger automated templates sent to MetaExperts API.

Recipient Number Sanitization:

- Trims non-digits.
- Appends "91" if length is exactly 10 digits.
- Discards invalid phone lengths (<11 or >15 digits).

- **Primary Contact Messages:**

- `Created` : Confirms receipt of order, invoice totals, and prepaid status.
- `Dispatched` : Sends tracking link, courier identifier, and estimated delivery date.
- `OFD` : Triggers Out for Delivery warning asking customer to stay reachable.
- `Delivered` : Confirms final payment validation or COD collection.
- `NDR` : Alerts the buyer of a failed delivery attempt and lists contact links.
- `Return/RDC` : Details tracking updates for packages returning back to origin (RTO).

- **Secondary Contact Messages:**

- Sends matching summaries to alternate contacts to ensure delivery rates remain high.

6. Live API Integrations

The OMS system supports live integration modes for **Delhivery** and **XpressBees** APIs. If API keys are set to 'MOCK' or mock credentials, the systems run simulated sandboxes.

6.1 Delhivery Integration

API endpoints connect to the staging (`staging-express.delhivery.com`) or production (`track.delhivery.com`) networks. 1. **AWB Allocation**: Requests a single waybill number from the waybill retrieval system: `GET /waybill/api/fetch/json/?token={{token}}&cl={{clientName}}&client_name={{clientName}}` 2. **Shipment Creation (CMU Manifest)**: Books the shipping order: `POST /api/cmu/create.json` Payload formatted as form url-encoded with JSON content: `format=json&data={ shipments: [...] }` 3. **Label Generation**: Fetches packing slip PDF data: `GET /api/p/packing_slip?wbns={{waybill}}` 4. **Tracking**: Fetches shipment logs: `GET /api/v1/packages/json/?token={{token}}&waybill={{waybill}}`

6.2 XpressBees Integration

1. **JWT Authentication**: Fetches a bearer token from: `POST /api/users/login` (body: `{ email, password }`). Token is cached for **12 hours** to optimize API usage.
 2. **Shipment Booking**: `POST /api/shipments2` (Payload: JSON containing order number, consignee data, pickup address, and dimensions).
 3. **Cancellation**: `POST /api/shipments2/cancel` (Payload: `{ awb: waybill }`).
 4. **Manifesting**: `POST /api/shipments2/manifest` (Payload: `{ awbs: [waybill] }`).
 5. **Reverse Pickup Booking**: `POST /api/Reverseshipments`.
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7. Local Development Guide

7.1 Setup Requirements

1. Install dependencies: `bash npm install`
2. Ensure local file read/write permissions are available for the `data/db.json` file. The database creates a backup copy (`db.json.tmp`) before writing atomically.

7.2 Running Development Server

Start the Next.js development server:

```
npm run dev
```

Open <http://localhost:3000> in your browser.

7.3 Testing Roles and Operations

- Login as `admin` to access the full system.
- Once logged in, look at the **Reviewer Bar** at the top of your screen to switch role context instantly and verify UI limits and permissions.
- Check the [Delhivery 99Store Manual Setup.md](#) for information on importing the workspace configuration into Postman.